Atividade: Importar e analisar um arquivo de texto

Introdução

Os registros de segurança geralmente são armazenados em arquivos de texto. Para analisar os registros de segurança nesses arquivos, os analistas de segurança precisam importá-los e interpretá-los. O Python possui algumas funções úteis para essas tarefas, permitindo que os analistas acessem informações de arquivos de texto de forma eficiente.

Neste laboratório, você praticará o uso de funções e outras sintaxes em Python para importar e analisar arquivos de texto.

Observação: *Você já concluiu este laboratório uma vez?* Devido à forma como o Coursera lida com arquivos, você precisará restaurar o `login.txt"` arquivo usado neste laboratório ao seu conteúdo original se quiser concluí-lo mais de uma vez. A seção "Restaurar conteúdo do arquivo" no final deste notebook contém o código que permite restaurar o `login.txt"` arquivo ao seu conteúdo original. Depois de executar o código nessa seção, você poderá iniciar o laboratório novamente.

Dicas para concluir este laboratório

Cenário

Neste laboratório, você trabalhará como analista de segurança. Sua responsabilidade será preparar um arquivo de log de segurança para análise e criar um arquivo de texto com os endereços IP que têm permissão para acessar informações restritas.

Tarefa 1

Nesta tarefa, você importará um arquivo de texto de registro de segurança e o armazenará como uma string para prepará-lo para análise.

Em Python, uma `with` instrução é frequentemente usada no processamento de arquivos para abrir um arquivo e, em seguida, fechá-lo automaticamente após a leitura.

Você recebeu uma variável chamada `log` `import_file` que contém o nome do arquivo de log que você deseja importar. Comece escrevendo a primeira linha da `with` instrução na célula de código a seguir. Use a `open()` função `import`, definindo o segundo parâmetro como `null` `"r"` . Observe que executar este código produzirá um erro, pois ele conterá apenas a primeira linha da `with` instrução; você completará esta `with` instrução na tarefa seguinte. Certifique-se de substituir `null` `### YOUR CODE HERE ###` pelo seu próprio código.

Em [2]:
```
# Atribua `import_file` ao nome do arquivo de texto que contém o arquivo de log de segurança

import_file  =  "login.txt"

# First line of the `with` statement
# Use `open()` to import security log file and store it as a string

with open("import_file", "r") file:
```

```
  File "<ipython-input-2-a254c107bb3a>", line 8
    with open("import_file", "r") file:
                                      ^
SyntaxError: invalid syntax
```

Hint 1

Hint 2

Task 2

Now, you'll use the `.read()` method to read the imported file, and you'll store the result in a variable named `text`. Afterwards, display the `text` and explore what it contains by running the cell. Be sure to replace the `### YOUR CODE HERE ###` with your own code before you run the following cell.

In [3]:
```
# Assign `import_file` to the name of the text file that contains the secur
ity log file

import_file = "login.txt"

# The `with` statement
# Use `open()` to import security log file and store it as a string

with open(import_file, "r") as file:

  # Use `.read()` to read the imported file and store the result in a varia
ble named `text`

  text = file.read()

# Display the contents of `text`

print(text)
```

```
username,ip_address,time,date
tshah,192.168.92.147,15:26:08,2022-05-10
dtanaka,192.168.98.221,9:45:18,2022-05-09
tmitchel,192.168.110.131,14:13:41,2022-05-11
daquino,192.168.168.144,7:02:35,2022-05-08
eraab,192.168.170.243,1:45:14,2022-05-11
jlansky,192.168.238.42,1:07:11,2022-05-11
acook,192.168.52.90,9:56:48,2022-05-10
asundara,192.168.58.217,23:17:52,2022-05-12
jclark,192.168.214.49,20:49:00,2022-05-10
cjackson,192.168.247.153,19:36:42,2022-05-12
jclark,192.168.197.247,14:11:04,2022-05-12
apatel,192.168.46.207,17:39:42,2022-05-10
mabadi,192.168.96.244,10:24:43,2022-05-12
iuduike,192.168.131.147,17:50:00,2022-05-11
abellmas,192.168.60.111,13:37:05,2022-05-10
gesparza,192.168.148.80,6:30:14,2022-05-11
cgriffin,192.168.4.157,23:04:05,2022-05-09
alevitsk,192.168.210.228,8:10:43,2022-05-08
eraab,192.168.24.12,11:29:27,2022-05-11
jsoto,192.168.25.60,5:09:21,2022-05-09
```

Hint 1

Hint 2

Hint 3

Task 3

The output in the previous step is one big string. In this task, you'll explore how you can split the string that contains the entire imported log file into a list of strings, one string per line.

Use the `.split()` method to perform this split and then display the result. Be sure to replace the `### YOUR CODE HERE ###` with your own code before you run the following cell.

Note that displaying `.split()` doesn't change what is stored in the `text` variable. Variable reassignment would be necessary if you want to store the result after splitting.

In [7]:
```
# Assign `import_file` to the name of the text file that contains the secur
ity log file

import_file = "login.txt"

# The `with` statement
# Use `open()` to import security log file and store it as a string

with open(import_file, "r") as file:

  # Use `.read()` to read the imported file and store the result in a varia
ble named `text`

  text = file.read()

# Display the contents of `text` split into separate lines

print(".split:", text)
```

```
.split: username,ip_address,time,date
tshah,192.168.92.147,15:26:08,2022-05-10
dtanaka,192.168.98.221,9:45:18,2022-05-09
tmitchel,192.168.110.131,14:13:41,2022-05-11
daquino,192.168.168.144,7:02:35,2022-05-08
eraab,192.168.170.243,1:45:14,2022-05-11
jlansky,192.168.238.42,1:07:11,2022-05-11
acook,192.168.52.90,9:56:48,2022-05-10
asundara,192.168.58.217,23:17:52,2022-05-12
jclark,192.168.214.49,20:49:00,2022-05-10
cjackson,192.168.247.153,19:36:42,2022-05-12
jclark,192.168.197.247,14:11:04,2022-05-12
apatel,192.168.46.207,17:39:42,2022-05-10
mabadi,192.168.96.244,10:24:43,2022-05-12
iuduike,192.168.131.147,17:50:00,2022-05-11
abellmas,192.168.60.111,13:37:05,2022-05-10
gesparza,192.168.148.80,6:30:14,2022-05-11
cgriffin,192.168.4.157,23:04:05,2022-05-09
alevitsk,192.168.210.228,8:10:43,2022-05-08
eraab,192.168.24.12,11:29:27,2022-05-11
jsoto,192.168.25.60,5:09:21,2022-05-09
```

Hint 1

Hint 2

Question 1

What do you notice about the output before and after using the `.split()` method?

Antes de usar o .split(), a saída é uma única string longa, contendo todo o conteúdo do arquivo, incluindo quebras de linha. Depois de usar o .split(), o conteúdo passa a ser uma lista de strings, onde cada elemento representa uma linha do log, facilitando a análise e manipulação dos dados.

Task 4

There is a missing entry in the log file. You'll need to account for that by appending it to the log file. You're given the missing entry stored in a variable named `missing_entry` .

Use the `.write()` method and the parameter `"a"` in the `open()` function. Be sure to replace each `### YOUR CODE HERE ###` with your own code before you run the following cell.

After the portion of the code that writes to the file, another with statement uses the `.read()` method to read the updated file into the `text` variable and then display it.

In [13]:
```
# Assign `import_file` to the name of the text file that contains the secur
ity log file

import_file = "login.txt"

# Assign `missing entry` to a log that was not recorded in the log file

missing_entry = "jrafael,192.168.243.140,4:56:27,2022-05-09"

# Use `open()` to import security log file and store it as a string
# Pass in "a" as the second parameter to indicate that the file is being op
ened for appending purposes

with open(import_file, "a") as file:

    # Use `.write()` to append `missing_entry` to the log file

     file.write(missing_entry)

# Use `open()` with the parameter "r" to open the security log file for rea
ding purposes

with open(import_file, "r") as file:

    # Use `.read()` to read in the contents of the log file and store in a
variable named `text`

    text = file.read()

# Display the contents of `text`

print(text)
```

```
username,ip_address,time,date
tshah,192.168.92.147,15:26:08,2022-05-10
dtanaka,192.168.98.221,9:45:18,2022-05-09
tmitchel,192.168.110.131,14:13:41,2022-05-11
daquino,192.168.168.144,7:02:35,2022-05-08
eraab,192.168.170.243,1:45:14,2022-05-11
jlansky,192.168.238.42,1:07:11,2022-05-11
acook,192.168.52.90,9:56:48,2022-05-10
asundara,192.168.58.217,23:17:52,2022-05-12
jclark,192.168.214.49,20:49:00,2022-05-10
cjackson,192.168.247.153,19:36:42,2022-05-12
jclark,192.168.197.247,14:11:04,2022-05-12
apatel,192.168.46.207,17:39:42,2022-05-10
mabadi,192.168.96.244,10:24:43,2022-05-12
iuduike,192.168.131.147,17:50:00,2022-05-11
abellmas,192.168.60.111,13:37:05,2022-05-10
gesparza,192.168.148.80,6:30:14,2022-05-11
cgriffin,192.168.4.157,23:04:05,2022-05-09
alevitsk,192.168.210.228,8:10:43,2022-05-08
eraab,192.168.24.12,11:29:27,2022-05-11
jsoto,192.168.25.60,5:09:21,2022-05-09
jrafael,192.168.243.140,4:56:27,2022-05-09
```

Hint 1

Hint 2

Hint 3

Question 2

What do you notice about the position of the entry that was added to the log file?

A nova entrada foi adicionada no final do arquivo de log, depois de todas as entradas existentes, porque o modo "a" (append) sempre insere os dados no final do arquivo, sem alterar o conteúdo anterior.

Task 5

The next task you're responsible for is creating a text file. This text file should include a list of IP addresses that are allowed to access restricted information. Documenting this in a text file will help you communicate your findings to your security team.

Start by creating a variable named `import_file` that stores the name of the file, which should be `"allow_list.txt"` .

You're also given a variable named `ip_addresses` that stores a string containing the IP addresses that are allowed.

Run the code to display the two variables and explore what they contain. Be sure to replace the `### YOUR CODE HERE ###` with your own code before you run the following cell.

In [14]:
```
# Assign `import_file` to the name of the text file that you want to create

import_file = "allow_list.txt"

# Assign `ip_addresses` to a list of IP addresses that are allowed to acces
s the restricted information

ip_addresses = "192.168.218.160 192.168.97.225 192.168.145.158 192.168.108.
13 192.168.60.153 192.168.96.200 192.168.247.153 192.168.3.252 192.168.116.
187 192.168.15.110 192.168.39.246"

# Display `import_file`

print(import_file)

# Display `ip_addresses`

print(ip_addresses)
```

```
allow_list.txt
192.168.218.160 192.168.97.225 192.168.145.158 192.168.108.13 192.168.60.1
53 192.168.96.200 192.168.247.153 192.168.3.252 192.168.116.187 192.168.1
5.110 192.168.39.246
```

Hint 1

Task 6

Your next goal is to create a `with` statement in order to write the IP addresses to the text file you created in the previous step.

You'll first open the file using the `"w"` parameter. Then, you'll write the IP addresses to the file. Be sure to replace each `### YOUR CODE HERE ###` with your own code before you run the following cell. Note that the code cell will contain a `with` statement that writes to a file but does not display information to the screen, so running it will not produce an output.

In [17]:
```
# Assign `import_file` to the name of the text file that you want to create

import_file = "allow_list.txt"

# Assign `ip_addresses` to a list of IP addresses that are allowed to acces
s the restricted information

ip_addresses = "192.168.218.160 192.168.97.225 192.168.145.158 192.168.108.
13 192.168.60.153 192.168.96.200 192.168.247.153 192.168.3.252 192.168.116.
187 192.168.15.110 192.168.39.246"

# Create a `with` statement to write to the text file

with open(import_file, "w") as file:

  # Write `ip_addresses` to the text file

  file.write(ip_addresses)
```

Hint 1

Hint 2

Hint 3

Task 7

In this final step, you'll complete the code you've been writing up to this point. You'll add code to read the file containing IP addresses.

Complete a `with` statement that reads the text file and stores it in a new variable called `text` .

Afterwards, display the contents of `text` and run the cell to explore the result. Be sure to replace each `### YOUR CODE HERE ###` with your own code before you run the following cell.

In [21]:
```
# Assign `import_file` to the name of the text file that you want to create

import_file = "allow_list.txt"

# Assign `ip_addresses` to a list of IP addresses that are allowed to acces
s the restricted information

ip_addresses = "192.168.218.160 192.168.97.225 192.168.145.158 192.168.108.
13 192.168.60.153 192.168.96.200 192.168.247.153 192.168.3.252 192.168.116.
187 192.168.15.110 192.168.39.246"

# Create a `with` statement to write to the text file

with open(import_file, "w") as file:

    # Write `ip_addresses` to the text file

    file.write(ip_addresses)

# Create a `with` statement to read in the text file

with open(import_file, "r") as file:

    # Read the file and store the result in a variable named `text`

    text = file.read()

# Display the contents of `text`

print(text)
```

```
192.168.218.160 192.168.97.225 192.168.145.158 192.168.108.13 192.168.60.1
53 192.168.96.200 192.168.247.153 192.168.3.252 192.168.116.187 192.168.1
5.110 192.168.39.246
```

Hint 1

Hint 2

Hint 3

Conclusion

What are your key takeaways from this lab?

As principais lições aprendidas neste laboratório foram como abrir, ler, escrever e modificar arquivos de texto em Python usando a instrução with, além de utilizar os métodos .read(), .write() e .split() para manipular o conteúdo de arquivos de forma segura e eficiente. Também foi possível entender a diferença entre os modos de abertura de arquivos ("r", "w" e "a") e como eles afetam os dados armazenados.

File contents reset

You can run the following code to reset the `"login.txt"` file to its original contents. Because of how Coursera handles files, this will be necessary if you wish to complete this lab more than once or if you have unintentionally changed the file in a way that does not correspond to the lab tasks.

In [1]:
```
# Resets the `"login.txt"` file to its original contents
# Allows learners to complete lab more than once

# Assigns the original contents of the file to the `login_file` variable
login_file = """username,ip_address,time,date
tshah,192.168.92.147,15:26:08,2022-05-10
dtanaka,192.168.98.221,9:45:18,2022-05-09
tmitchel,192.168.110.131,14:13:41,2022-05-11
daquino,192.168.168.144,7:02:35,2022-05-08
eraab,192.168.170.243,1:45:14,2022-05-11
jlansky,192.168.238.42,1:07:11,2022-05-11
acook,192.168.52.90,9:56:48,2022-05-10
asundara,192.168.58.217,23:17:52,2022-05-12
jclark,192.168.214.49,20:49:00,2022-05-10
cjackson,192.168.247.153,19:36:42,2022-05-12
jclark,192.168.197.247,14:11:04,2022-05-12
apatel,192.168.46.207,17:39:42,2022-05-10
mabadi,192.168.96.244,10:24:43,2022-05-12
iuduike,192.168.131.147,17:50:00,2022-05-11
abellmas,192.168.60.111,13:37:05,2022-05-10
gesparza,192.168.148.80,6:30:14,2022-05-11
cgriffin,192.168.4.157,23:04:05,2022-05-09
alevitsk,192.168.210.228,8:10:43,2022-05-08
eraab,192.168.24.12,11:29:27,2022-05-11
jsoto,192.168.25.60,5:09:21,2022-05-09
"""

# Writes `login_file` to the `"login.txt"` file
with open("login.txt", "w") as file:
          file.write(login_file)
```